INTERNATIONAL UNIVERSITY OF APPLIED SCIENCES

Advanced Statistics - DLMDSAS01

University of Applied Science - Online

Study-branch: Data Science

Advanced Statistics Workbook

Abel Yann Djabou Nghongnang

Matrikelnummer: IU14096047

Fritz-Schröder-Str 27a

realized in Sankt Augustin

Advisor: Paul Libbrecht

Realized with input of the Parameter Generator: 0e3da8e25791c53bdd8cf393dee8eb2985211ad9

Delivery date: 6.11.2026

Task 1: Basic Probabilities and Visualizations (1)

The parameter generator provided the following values:

$$\xi_1 = 0, \qquad \xi_2 = 0.62.$$

In this task, a voter must choose one of two possible answers: either “for” or “against”. Before the vote takes place, we do not know which answer will be selected, but the probability model specified in the assignment allows us to describe the uncertainty mathematically.

A Bernoulli random variable represents a single trial with two possible outcomes, conventionally called success and failure (ref).

To describe the vote numerically, we define the random variable X by

$$X = \begin{cases} 1, & \text{if the vote is for,} \\ 0, & \text{if the vote is against.} \end{cases}$$

This definition makes a vote in favour the “success” outcome and a vote against the “failure” outcome. The numerical encoding is a modelling choice; it does not change the voting situation itself.

By the definition of a Bernoulli distribution, its parameter p is the probability of success, with $0 \leq p \leq 1$. The task specifies that the probability of voting “for” equals ξ_2. Consequently,

$$X \sim \mathrm{B}(p), \qquad p = \xi_2 = 0.62.$$

It follows directly that

$$P(X = 1) = p = 0.62.$$

Since the only other possible outcome is $X = 0$, its probability is the complement of the probability of voting “for”:

$$\begin{aligned} P(X = 0) &= 1 - P(X = 1) \\ &= 1 - 0.62 \\ &= 0.38. \end{aligned}$$

Thus, the probability model assigns probability 0.62 to a vote in favour and probability 0.38 to a vote against.

Interpreting and visualizing the probabilities

The probabilities can be expressed as percentages by multiplying each probability by 100%. Therefore,

$$P(\text{vote} = \text{“for”}) = 0.62 \cdot 100\% = 62\%$$

and

$$P(\text{vote} = \text{“against”}) = 0.38 \cdot 100\% = 38\%.$$

These percentages describe the probabilities specified by the model. They do not mean that a single vote is

partly “for” and partly “against”: the outcome of the individual trial is one or the other. The percentages quantify the model’s uncertainty before the outcome is observed.

A bar chart is an appropriate visualization because there are two discrete outcomes, each associated with a probability. Figure 1 displays the probabilities, with the voting outcomes on the horizontal axis and probability in percent on the vertical axis. The vertical scale runs from 0% to 100%.

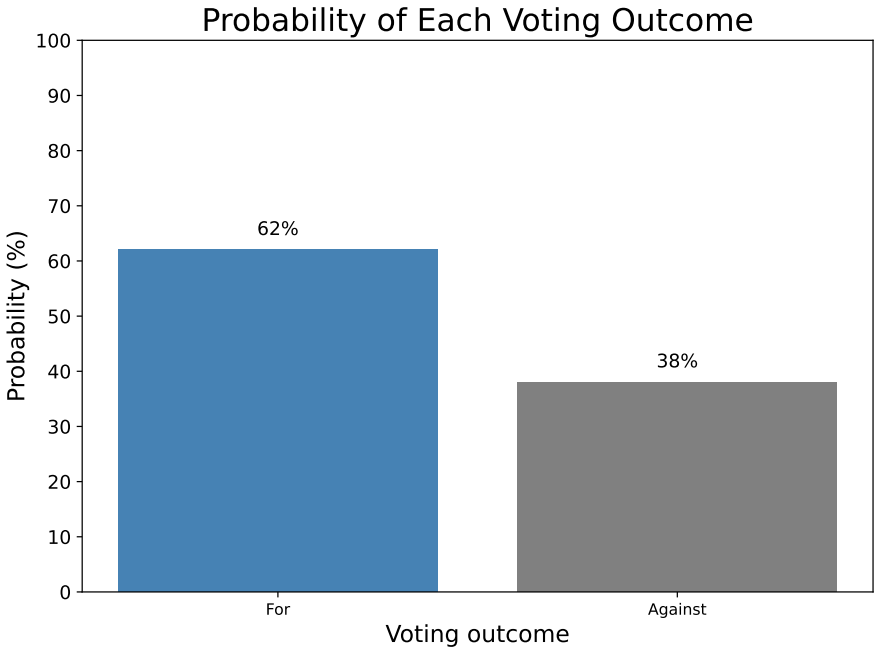


Figure 1: Probability distribution for a single vote, using $\xi_2 = 0.62$ (code provided below)

Before accepting these values as a valid probability distribution, we can verify the two basic requirements. First, neither probability is negative:

$$0 \leq 0.38 \leq 1, \qquad 0 \leq 0.62 \leq 1.$$

Second, because the two outcomes are mutually exclusive and exhaust all possibilities for this vote, their probabilities must sum to one. Indeed,

$$P(X = 0) + P(X = 1) = 0.38 + 0.62 = 1.$$

Both requirements are satisfied. Hence, the calculated probabilities form a valid distribution for the random variable X.

The figure is generated from these calculated values rather than from simulated votes. This distinction matters: simulation would produce random observed proportions, whereas the bars in Figure 1 display the probabilities specified by the model.

Can the expected value be calculated?

The next question is whether the random outcome has an expectation. To answer it, we must distinguish the original categorical answers from the numerical random variable X defined above. The words “for” and “against” do not, by themselves, have numerical values whose average can be calculated. Our definition of X, however, assigns the values 1 and 0 to these outcomes, so its expectation is mathematically meaningful.

For a discrete random variable, the expectation is defined as the sum of each possible value multiplied by its probability. When the random variable has finitely many possible values, this sum is finite and the expectation exists (ref). In this exercise, the definition gives

$$E[X] = \sum_{x \in \{0,1\}} xP(X = x).$$

Substituting the two possible values and their probabilities yields

$$\begin{aligned} E[X] &= 0 \cdot P(X = 0) + 1 \cdot P(X = 1) \\ &= 0 \cdot 0.38 + 1 \cdot 0.62 \\ &= 0 + 0.62 \\ &= \boxed{0.62}. \end{aligned}$$

For completeness, the existence of the expectation can also be verified using the sufficient condition $E[|X|] < \infty$. Since X can take only the values 0 and 1, we obtain

$$\begin{aligned} E[|X|] &= |0|P(X = 0) + |1|P(X = 1) \\ &= 0 \cdot 0.38 + 1 \cdot 0.62 \\ &= 0.62 < \infty. \end{aligned}$$

Therefore, $E[|X|]$ is finite, which establishes that the expectation of X exists and is finite.

The result has a straightforward interpretation. The random variable X equals one when the voter votes “for” and zero otherwise. Consequently, its expected value equals the probability of voting “for”:

$$E[X] = P(X = 1) = 0.62.$$

In repeated comparable voting situations, the expected value describes the long-run average of the numerical indicator, not the outcome of the individual vote. It does not imply that a particular vote will equal 0.62; an observed value of X must be either zero or one.

The numerical encoding should also be kept in mind. If we had instead defined $X = 1$ for “against” and $X = 0$ for “for”, the expectation would have been 0.38. The probabilities of the actual voting outcomes would remain unchanged. Thus, an expectation for categorical outcomes depends on the numerical representation chosen, and that representation must be stated explicitly.

```
1  # Import required libraries
2  import matplotlib.pyplot as plt
3
4  # Define the data
5  votes = ["For", "Against"]
6  probabilities = [62, 38]
7
8  # Create the figure
9  plt.figure(figsize=(8, 6))
10
11 # Draw the bar chart
12 bars = plt.bar(votes, probabilities, color=['steelblue', 'grey'])
13
```

```
# Beautifying the plot
plt.title('Probability of Each Voting Outcome', fontsize=20)
plt.xlabel('Voting outcome', fontsize=15)
plt.ylabel('Probability (%)', fontsize=15)

# Set axis limits and ticks explicitly
plt.ylim(0, 100)
plt.yticks(range(0, 101, 10), fontsize=12)

# Add the percentage labels on top of the bars
for bar, probability in zip(bars, probabilities):
    plt.text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height() + 2,
        f"{probability}%",
        ha="center",
        va="bottom",
        fontsize=12
    )

# Save and show
plt.tight_layout()
plt.savefig("pics/task1_bernoulli.pdf", dpi=300)
plt.show()
```

Task 2: Basic Probabilities and Visualizations (2)

0.1 Tools

MiKTeX: https://miktex.org/download TeXLive: http://tug.org/texlive/ (or alternative LaTeX-systems).

A good editor is essential. Sometimes combined editors and compilers (e.g. TeXShop) can be really productive. Make sure you learn the use of synchronized navigation then.

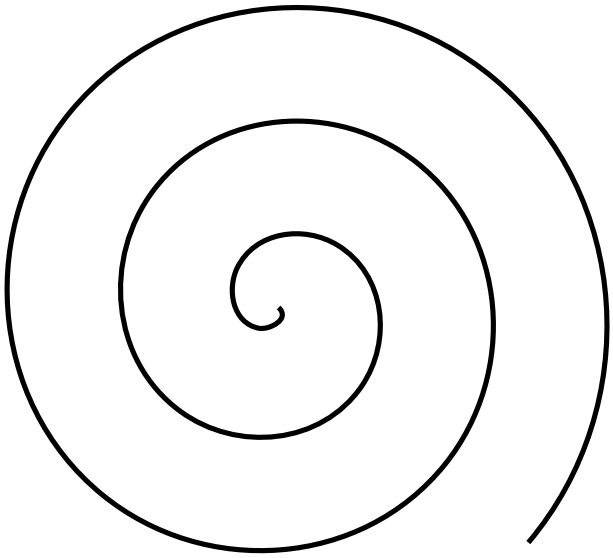


Figure 2: A spiral... smooth vector-based with a clean parametrisation!
Nothing to do with Gage (2018)

0.2 Tables

" Industrial era "	"Jobs "	" Wanted: Upgrade"'
Parts exchanger	Fitter	mecatronics specialist
eShop	reseller	" Client-suggester"
" Coding-guru"'	Softwaredesign	Whole-life designer
JA! Gut & Günstig	brand-names	" Life-Style Feeling"
Internetbanking	Bank clerk	Customer adviser
Robots	Specialist	Machine supervisor
Bush	Gardener	Nature-sculptor
Painting	Painter	Interior Design

Table 1: Downgrade and Upgrade of job denominations
Dueck (2016)

0.3 Listes

- one

- twoi
- threei

1. first
2. second
3. third

0.4 Formulæ

A formula can be written inline, e.g. as $\frac{d}{dx}\text{arctg}(x) = \frac{1}{1+x^2}$ or, in centered math:

$$\frac{d}{dx}\text{arctg}(x) = \frac{1}{1+x^2} \tag{0.1}$$

Notice that formulæ that are centered start bigger (technically, they start in `\displaystyle`) than they start inline (technically, they start in `\textstyle` all subsequents reductions, e.g. an exponent, goes to `\scriptstyle` then `\scriptscriptstyle`). Indeed a best effort is made so that inline formulæ do not change the line height which would bother the eye of a reader.

Formulæ can be given a number and a label. Numbering happens automatically with `\begin{equation}` and `\end{equation}` and can be avoided if enclosing the formula between `\[` and `\]`. If using the `\label` macro inside, you can refer automatically to this equation using `\ref{label}`. E.g. Thanks to equation 0.1 one dare say that:

$$\int_0^t \frac{1}{1+x^2}dx = \arctan(t) \tag{0.2}$$

0.5 Tools and Code

Many users of this template will want to include some code.

The simplest way to do so is to use the `\verb` macro which is followed by a sign, then some code, then the sign again to close. This is the inline version which works as in:

```
As we could calculate with \textcite{Wolfram_Alpha} using
\verb_integrate 1 / pi e ^ (t/pi) from zero to infinity_.
```

which yields:

As we could calculate with Wolfram Inc. (2021) using `integrate 1 / pi e ^ (t/pi)` from zero to infinity.

The multiline version of this is called `\begin{verbatim}` and finishes with `\end{verbatim}`.

0.6 Citation examples

Monography (Liebel, 2019, p. 22)

Collection (Müller & Mustermann, 2019)

Article (Schülers, 2021)

Bibliography

Dueck, G. (2016). Zwischen industralisierendem Downgrade und notwendigem Upgrade/Empowerment [auf dem AKAC-Kongress].

Gage, J. (2018). Introduction to Microservices [Accessed: 02.06.2019]. https://blog.algorithmia.com/introduction-to-microservices/

Liebel, O. (2019). *Skalierbare Container-Infrastrukturen*. Rheinwerk Verlag.

Müller, P., & Mustermann, M. (2019). Artikel in Sammelband [Sammelband für IUBH]. Verlag.

Schülers. (2021). Zeitschriftenartikel [In: Frankfurter Allgemeine Zeitung], *10.* p. 10 –14.

Wolfram Inc. (2021). Wolfram Alpha [Accessed: 10.06.2021]. https://wolframalpha.com

INTERNATIONAL
UNIVERSITY OF
APPLIED SCIENCES

Eidesstattliche Erklärung

I hereby certify...

.....	.....
Place, date	Signature